

Cybersecurity Concepts Lab for Teachers

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FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-in-inked, in the order the lessons take them.

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OCTOBER 2026

Demo 7: Updates and "Patch the Hole" (The Fence Pattern Game)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE	
Big idea	Software has mistakes. Some mistakes are holes that can be misused. An update is a fix for a known hole. The race is between the fix reaching your device and someone using the hole.
Grade bands	K-2 (fix the fence), 3-5 (full pattern game), 6-8 (full + spread dice), 9-12 (full + zero-day vs patched, update tradeoffs, supply chain)
CSTA-style concepts	Computing Systems (troubleshooting, hardware and software), Networks and the Internet (cybersecurity), Algorithms and Programming (patterns, debugging) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed fence strips, colored pencils or stickers, dice. No devices.
Teachers do it first	Yes. Part A about 20 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Find the hole in this fence: R B B R B B R B R R B B. Fix it and say "That is a patch." Then ask: if this fence went around a whole town, how many fences need fixing? (Every copy. That is why updates must reach every device.)

01 The big idea

EVERY PROGRAM IS WRITTEN by people, and people make mistakes. Most mistakes are harmless (a button in the wrong place). A few are **security holes** (called **vulnerabilities**): a gap that someone could misuse to get in, read things or take control. When a hole is found, the maker writes a **patch** (a fix) and ships it as an **update**.

The updating race has three clocks:

1. **Discovery.** Someone finds the hole (a researcher, a user, the maker, or a wrongdoer).
2. **Fix.** The maker writes and releases a patch.
3. **Install.** Everyone must actually install the patch.

Most real-world harm happens in the gap between "a fix exists" and "you installed it", because once a fix is published, wrongdoers can study it to learn what the hole was. Your job as a defender is to **shrink the third clock: turn on automatic updates**.

02 What a teacher should know (plain language)

2.1 VOCABULARY

TERM	PLAIN MEANING
Bug	A mistake in software.
Vulnerability	A bug (or design weakness) that could be misused to break security.
Exploit	The method used to take advantage of a vulnerability. (We do not teach how to write one.)
Patch / update	The fix.
Zero-day	A hole nobody on the defense side knew about, so "zero days" to fix it before it was used. Rare and valuable.
N-day (known vulnerability)	A hole that has a published fix but is still unpatched on some devices. Far more common in real incidents.
End-of-life (EOL)	A product that no longer gets updates; its holes stay open forever.
Auto-update	The device installs patches by itself.

2.2 WHY UPDATES FEEL ANNOYING AND WHY WE DO THEM ANYWAY

- They interrupt us at bad times (restarts, progress bars).
- A small number of updates have caused new problems. Big organizations test before rolling out; **home and school users should usually still prefer prompt automatic updates**, as the risk from known

holes is usually larger than the risk from a bad update [VERIFY consensus guidance from CISA / NCSC].

- Updates also bring features, battery fixes and compatibility, so the "just cosmetic" myth is wrong.

2.3 TRUE STORIES (FOR TEACHERS; KEEP STUDENT VERSIONS GENTLE)

- **WannaCry (2017).** A worm used a Windows hole for which Microsoft had released a patch about two months earlier; machines that were not patched (or were no longer supported) were hit, including large organizations such as parts of the UK National Health Service [VERIFIED 2026-10-05: Microsoft Security Bulletin MS17-010 was published March 14, 2017; the UK National Audit Office says the attack hit the NHS on 12 May 2017, that NHS Digital had issued critical alerts to patch in March and April 2017, and that affected organisations had unpatched or unsupported Windows systems; at least 34% of trusts in England were disrupted].
- **Equifax (2017).** A breach exposing the personal data of roughly 147 million people was linked to an unpatched vulnerability in a web framework (Apache Struts) for which a fix had been available [VERIFIED 2026-10-05: FTC complaint against Equifax: more than 147 million consumers; US-CERT alerted Equifax to the Apache Struts flaw on March 8, 2017 and the patch was available; Equifax failed to patch the affected portal for months].
- Both illustrate the "N-day" lesson: **a patch existed.**

2.4 WHAT TO UPDATE (THE CHECKLIST)

- Phone and tablet operating system; computer operating system.
- Browser (and browser extensions).
- Apps and games.
- Home router (the box that connects the house to the internet; many do not update themselves) [VERIFY by model].
- Smart devices (TVs, cameras, toys, speakers).
- Replace devices that are **end-of-life**, or isolate them. Do not keep using an unsupported device for sensitive things.

2.5 UPDATE SCAMS (TIE TO DEMO 3)

FAKE "UPDATE" POP-UPS ARE a common trick: "Your video player is out of date, click to update" on a random website. **Rule:** install updates only from the device's own settings or the official app store, never from a pop-up on a web page or a link in a message.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Updates only change the look."	Many are security fixes.
"My phone/Mac/Chromebook can't get hacked."	Less common is not never; any software has bugs.
"Wait for others to try it first."	OK for big organizations, not for home use for security patches.
"Antivirus replaces updates."	Antivirus helps; it does not close the hole.
"Updates are a way to track me."	Updates come from the maker; be careful only with updates from non-official sources.

03 Materials (zero to low cost)

- **Fence Strips** (Section 4.1; 12 to 16 slats each, on grid paper with colored pencils or stickers).
- **Village cards:** 20 index cards, each a "house" with "v1" (unpatched) on one side and "v2" (patched) on the other. Two six-sided dice per group.
- A "town crier" bell (any noise-maker).
- A scoreboard on the board.
- Student handout.

04 The Fence Pattern Game

4.1 WARM-UP: FIND THE HOLE (PATTERN GAME)

A FENCE FOLLOWS A repeating pattern. A **hole** is where the pattern breaks. In real software, holes are not so visible. They are discovered by testing, by researchers, by users reporting issues, and by the maker's own checks. For the game, we use pattern-breaking as a stand-in for "something is wrong here".

Strip 1 (pattern RBB, 12 slats): R B B R B B R B R B B B

Strip 2 (pattern RRGB, 12 slats): R R G B R R G B R G G B

Strip 3 (pattern 1 2 3 3, 16 slats): 1 2 3 3 1 2 3 3 1 2 3 3 1 2 3 1

Strip 4 (pattern ABAB, 12 slats, two holes): A B A B A B B A B A A

Answers are on the teacher answer page.

Students fill each hole by rewriting the correct slat. Say: **"That is a patch."** Then ask: "If this fence went around a whole town, how many fences need fixing?" (Every copy. That is why updates must reach every device.)

4.2 SPREAD DICE: "GREMLIN VILLAGE"

This is a model that uses a cartoon "gremlin" (a stand-in for any misuse of a known hole) and **no real malware details**.

Setup: 20 house cards in a 4 by 5 grid. All start as v1 (unpatched, "holey"). One house, chosen by the teacher, starts with a gremlin token.

Rules per round:

1. For each house **next to** a house with a gremlin (up, down, left, right), roll one die. On **1, 2 or 3** a gremlin gets in **only if the house is v1**. A v2 (patched) house is **immune**. Flipping a house to v2 does **not** remove a gremlin already inside: a patch closes the hole, but cleaning up someone already in takes extra work (a real-world point worth saying aloud).
2. Count houses with gremlins at the end of the round.
3. After round 2, the Town Crier rings: **"A patch is ready!"** Then the game changes depending on scenario.

Scenarios (run in different groups, then compare):

SCENARIO	AFTER THE BELL
A: "No one updates"	Nothing changes. Play four more rounds.
B: "Some update"	Each house rolls a die: 1-3 flip to v2 now, 4-6 stay v1.
C: "Auto-update"	All houses flip to v2 immediately. Gremlins stop spreading (but those already in stay).
D: "Late update"	Wait one full extra round before flipping all to v2.

Record counts each round. Compare the graphs (draw a simple bar chart or a line). The class sees that **time to patch** and **how many patch** both matter, and that a patched neighbor protects you a bit by not passing the gremlin along.

Teacher note on honesty: real spread depends on many things (networks, what the problem actually does). This is a toy that shows a pattern. Say so.

4.3 ZERO-DAY CARD (6-12): "THE HOLE NOBODY KNOWS"

Give each group a hole card that says "Nobody has told the maker yet." Ask: can any house be protected by patching? (No: no patch exists.) What can defenders still do? (Limit what each house can do, watch for strange behavior, keep backups (Demo 10), stay alert to phishing (Demo 3), report problems quickly.) Defense in depth: no single layer is perfect.

05 Part A: Teachers do it first (about 20 minutes)

1. **Pattern warm-up (5 min).** Solve Strips 1-4. The brain is good at spotting patterns and bad at spotting small errors in long ones.
2. **Gremlin Village (10 min).** Play Scenarios A and C, rolling honestly and recording totals. Predict B and D before rolling.
3. **Reflect (5 min).** Which scenario did best, and which is most like your own devices? Check which of your devices are on **auto-update** and which are end-of-life.

06 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook.
2-6	Fence strips: students find holes in strips 1-3 and "patch".
6-12	Gremlin Village: one class-wide game, Scenario A for two rounds, then the bell, then Scenario C. Record counts on the board.
12-15	Debrief and exit slip: "Turn on _____ updates."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-4	Hook and vocabulary.
4-12	Fence strips including the two-hole Strip 4; relate to debugging.
12-30	Gremlin Village: groups of four run all four scenarios (rotating roles: roller, recorder, house-flipper, graph-maker).
30-38	Zero-day card, then the "update checklist" sort: students classify cards (phone OS, router, old tablet, fake pop-up, game) into "update now", "replace", "ignore this".
38-43	Update scams: spot the fake update message among real ones.
43-45	Exit ticket and handout.

07 Teacher script

Hook (2 minutes). "A fence goes around the town. Somewhere there's a slat that doesn't follow the pattern. Can you find it? You just did what security researchers do when they hunt for mistakes. Now let's see what happens when a hole in a whole town's fence is found and a fix arrives. Does everyone fix it at the same time?"

After the bell. "Notice what we did? The fix existed. Houses that stayed v1 stayed open. That is the real-world story, and it is why the best habit you can have is a boring one: automatic updates."

On zero-days. "What if the hole is a secret, with no patch? Then we use other layers: limit what each thing can do, keep backups, stay alert to tricks, report weirdness. No single layer wins."

Close. "Updates are like a vaccine for your devices: not perfect, but a big advantage. Turn on automatic updates tonight with a grown-up. And never install an 'update' from a pop-up. Pause, check, tell."

08 Kid-friendly analogies by band

BAND ANALOGY AND ACTIVITY

K-2	Fix the fence. A fence with a missing board; a puppy (a cartoon, never real danger) could get out. We fix the board and tell the grown-up. "Say yes to updates when a grown-up says it's safe."
3-5	Pattern spotting and the town bell. Strips 1-3 plus a mini Gremlin Village with 12 houses.
6-8	Spread dice. Full Village, graphs, and scenarios. Update scams and the checklist.
9-12	Time-to-patch economics. Zero-day vs n-day, supply-chain and end-of-life risk, enterprise patch testing, why automatic updates are a safety feature and what tradeoffs exist, vulnerability databases (CVE) and severity scoring.

09 Discussion questions

1. Why is "the fix exists" not the same as "I am safe"?
2. In the village, why did patched houses help their neighbors?
3. Why do scammers pretend to be updates?
4. (9-12) Why might a company delay a patch for a few days, and what is the cost of waiting?

10 Level-up challenge

- **Design a patch plan** for a fictional school with 500 devices: which update first (Internet-facing, administrators), how to test, how to communicate, how to handle old devices.
- **Read a real advisory** (a public "security update" note from a major vendor): what is fixed, how serious, who needs to act [VERIFY a suitable public advisory; do not discuss exploit details].
- **Software supply chain (11-12).** How a trusted update channel itself can be attacked and why signed updates matter [VERIFY with a public case such as SolarWinds 2020].

11 Safety and ethics notes (defense only)

- Never teach, write or run exploit code. The "gremlin" is a cartoon token. Do not describe how any specific hole is misused.
- Do not scan, probe or test any real device, router or website for out-of-date software, including at home or school, without explicit permission from the owner. Checking your *own* device's update screen is fine.
- Do not encourage students to change settings on school-managed devices. Updates on school devices are IT's job.
- Remind students to ask a trusted adult before updating family devices and never to download "updates" from links.
- If a student says someone asked them to install something on their device, treat it as a possible safety concern and tell a trusted adult.

12 Assessment idea

- **Exit ticket.** "Name two things to update and one place an update should NOT come from." Rubric: 3 = two devices/apps, plus pop-ups or links; 2 = one device and some source; 1 = "turn off updates".
- **Pattern transfer.** Provide a new strip; students find holes and explain "that is a patch for one copy; real updates must reach every copy".

13 Sources and [VERIFY] list

- Microsoft Security Bulletin MS17-010 (March 14, 2017) and UK NAO report on WannaCry and the NHS (27 Oct 2017) [VERIFIED 2026-10-05: learn.microsoft.com; nao.org.uk].
- FTC complaint (2019) and US GAO report GAO-18-559 on the Equifax 2017 breach [VERIFIED 2026-10-05: ftc.gov complaint read; GAO report located]; the House Oversight report was not read.
- CISA Known Exploited Vulnerabilities catalog (a CISA list of vulnerabilities known to be exploited) [catalog page exists: VERIFIED 2026-10-05: cisa.gov].
- NCSC (UK) guidance on keeping devices up to date [not checked; the page tried was not readable].
- CVE program and CVSS severity scoring (MITRE, FIRST) [VERIFY].
- Microsoft "Patch Tuesday" monthly release schedule [VERIFY].

Student Handout: Patch the Hole

Name: _____

Words to know

- **Bug:** a mistake in software.
- **Vulnerability:** a bug someone could misuse.
- **Patch / update:** the fix.
- **Auto-update:** the device installs fixes by itself.

Find the hole! Write the missing slat.

1. R B B R B B R B R R B B The hole is slat ____ . It should be ____ .
2. A B A B A B B B A B A A The holes are slats ____ and ____ .

Gremlin Village results

ROUND	SCENARIO A: NO ONE UPDATES	SCENARIO C: AUTO-UPDATE
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____

WHICH SCENARIO KEPT THE most houses safe? ____ Why?

My update checklist (with a grown-up)

- ☐ Phone or tablet: automatic updates ON
- ☐ Computer: automatic updates ON
- ☐ Browser and apps: updated
- ☐ Game console or smart device: updated
- ☐ Old device that no longer gets updates? Ask a grown-up what to do.

Real or fake update?

A **POP-UP ON A** website says: "Your video player is out of date! Click to update." Real or fake? _____ What do I do? _____

Remember

- Updates fix holes. The fix only helps when it is **installed**.
- Install updates only from your device's own settings or the official app store.
- If someone asks you to download something to "fix" your device, **Pause. Check. Tell.**

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 7 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Fence strips (verified): Strip 1 (pattern RBB): slat 9 should be B. Strip 2 (pattern RRGB): slat 10 should be R. Strip 3 (pattern 1 2 3 3): slat 16 should be 3. Strip 4 (pattern ABAB, two holes): slat 7 should be A and slat 12 should be B.

Handout: strip 1, slat 9 should be B; strip 4, slats 7 (A) and 12 (B).

Gremlin Village, typical results. 4 by 5 grid, one starting gremlin near the middle, 50 percent chance per exposed v1 house per round, bell after round 2. Average houses with a gremlin at the end of rounds 1 to 6 (simulated over 20,000 games; your dice will vary widely):

SCENARIO	R1	R2	R3	R4	R5	R6
A: no one updates	3	6	10	14	16	18
B: some update (half)	3	6	8	10	11	11
C: auto-update	3	6	6	6	6	6
D: late update (one extra round)	3	6	10	10	10	10

Round 1 includes the starting house. Houses already holding a gremlin stay that way after patching. Auto-update (C) keeps the most houses safe; D shows that waiting one round cost about four more houses.

Real or fake update? Fake. Close it and tell an adult; install updates only from device settings or the official app store.